VULCAN

Describe it. Hold it.

Product Specification & Solo-Founder Build Plan · v0.2 · July 2026 · Prepared for Ethan Bernstein · Confidential

Vulcan turns a photo and a sentence into a physical object delivered to your door. It is the store for things that don't exist: the bracket for your weird shelf, the knob discontinued in 1994, the mount that makes a standard product fit your specific wall. The company is API-first software — an intent-to-part engine — with fabrication outsourced from day one, except for Phase 0, where the founder's own 3D printers fulfill orders to validate demand and generate revenue with near-zero capital.

Secondary tagline (ads / packaging): **“They don't make that. We do.”**

1 Product Overview

1.1 The problem

Mass production means you can only buy what already exists. Custom fabrication exists (Xometry, Protolabs, local shops) but is gated behind CAD literacy, which excludes ~99% of people. The moment Vulcan serves is universal and unserved: **“they don't make that.”**

1.2 Product definition

A consumer web product (thin client) on top of the **Vulcan Core API**. The user submits a photo of the context plus a plain-language description. Vulcan infers dimensions, synthesizes a manufacturable design, verifies it for the target process, shows a render and a price, and ships the object in 2–5 days. Every delivered part generates a fit-outcome record — the proprietary data asset.

1.3 Who it's for

Persona	Trigger moment	Willingness to pay
Home fixer	Broken/missing part; appliance knob, hinge, bracket, cap	$15–60 to avoid replacing a $300 appliance
Space adapter (RV, van, boat, small apartment)	Nothing standard fits the odd space	$25–120; community-driven, high word of mouth
Hobbyist / niche gear owner	Adapter or mount between two products that were never designed to meet	$20–80; repeat-heavy segment
Small business (Phase 1+)	Jigs, fixtures, spare parts for orphaned equipment	$50–500; becomes the B2B API wedge

1.4 Scope — v1 wedge

In scope (v1 categories)	Examples	Process / materials	Price band
Brackets & mounts	Shelf brackets, wall mounts, camera/device mounts	FDM · PETG, PLA, CF-PETG	$19–79
Adapters & couplers	Hose, vacuum, tripod, pole, tube adapters	FDM · PETG, TPU	$15–49
Enclosures & trays	Boxes, organizers, drawer inserts, device cases	FDM · PLA, PETG	$19–69
Knobs, handles, caps	Appliance knobs, replacement handles, end caps	FDM · PETG, TPU	$12–39
Clips & hooks	Rail hooks, cable clips, hangers	FDM · PETG, TPU	$9–29

Explicitly out of scope (v1): load-critical or safety parts (climbing, automotive suspension, child seats), pressure vessels, food-contact items, anything regulated (medical, aerospace), electronics, optics, parts >250×250×250 mm, orders >20 identical units (route to molding partners later). Out-of-scope requests get an honest “we can't make that (yet)” + a link if the thing exists + the request is logged as demand data.

1.5 Positioning

Player	What they sell	What Vulcan does differently
Xometry / Protolabs	Manufacturing for people with CAD files	No file required; Vulcan originates the design from intent
Backflip AI, Zoo (text-to-CAD)	Design tools / APIs for people who think in parts	Vulcan sells the delivered object, not a tool; owns fit-outcome data
Etsy / eBay sellers	Pre-designed printed items	Per-customer geometry, engineered fit, guaranteed dimensions
Amazon	Things that already exist	Things that don't

2 How It Operates — the Order Pipeline

#	Stage	What happens	Powered by
1	Intent capture	Photo(s) + plain-language request via web app	Frontier LLM w/ structured output → canonical IntentSpec
2	Dimension & scale resolution	Monocular metric-depth prior sizes the scene; guided-measure prompts confirm every critical dimension (see §2.2)	Depth Pro-class metric depth model + guided-measure UI; per-dim source + confidence tags
3	Design synthesis	Track A (default): parametric template filled with inferred dims. Track B (fallback): freeform text-to-CAD, founder-reviewed	CadQuery/build123d template engine; Zoo API adapter for Track B
4	DFM validation	Manifold check, min wall, overhangs, clearances/tolerances for process+material, printability sim	trimesh/manifold3d + slicer CLI (PrusaSlicer) as oracle
5	Preview & confirm	Render + key dimensions + material + price; user approves	Headless render; quote from pricing engine
6	Pricing & quote	Price = f(material g, machine min, labor, ship, margin floor)	Slicer output drives cost model (§5)
7	Fulfillment routing	Phase 0: founder's printers. Phase 1+: partner farms via API, CNC/sheet-metal shops for metal	Order queue → gcode pack per printer profile
8	QA & ship	Dimensional spot-check against spec card; pack; label	Per-order QA checklist auto-generated from IntentSpec
9	Fit-outcome capture	“Did it fit?” flow (photo of part in place = $5 credit); returns & reprints logged	Outcome rows → the data flywheel

2.1 Why dual-track design synthesis

Free-form text-to-CAD is not yet reliable enough to ship unreviewed. Track A (parametric templates) covers ~80% of wedge demand with near-zero failure: each category is a hand-built, DFM-safe parametric model with 5–15 parameters the LLM fills from the photo and conversation. Track B (freeform generation) serves the long tail with founder review of every design before printing — review time is the price of learning what to templatize next. The ratio shifts toward Track B as models and outcome data mature.

2.2 Scale resolution — guided measurement + metric-depth prior

A photo has no absolute scale, and ±2 mm is the difference between a part and a return. Vulcan v1 resolves scale with two cooperating sources; every dimension in a design carries a **source tag** and a **confidence score**, and orders cannot be placed while any fit-critical dimension remains unconfirmed.

Source	How it works	Trust policy
Tier 2 — Guided manual measurement	The API returns questions[] rendered as arrows drawn on the user's own photo: “measure this gap,” “what is the diameter of this pole?” User types the number (mm or inches).	Sole acceptable source for fit-critical dims (shaft diameters, gaps, hole spacing, anything ±1mm). Tag: user_measured.

Source	How it works	Trust policy
Tier 3 — Monocular metric depth	A Depth Pro / Metric3D-class model + EXIF focal length produces absolute-scale scene depth, sizing all non-critical dimensions and proposing defaults (typically within a few %).	Auto-fills cosmetic/loose dims (tag: depth_inferred). Never sole source for critical dims — used to propose, not to commit.
Cross-check (2 ⊗ 3)	Every user-typed value is compared against the depth prior; disagreement >20% triggers a re-ask ("you entered 25 cm — the photo suggests ~2.5 cm — please double-check").	Catches unit mistakes (mm/cm/in), the single most common measurement error. Mismatch always re-asks; never silently overrides the user.

Design-side hedge: templates absorb residual uncertainty wherever fit allows — slotted holes instead of round, compliant snap features, clamps with adjustment range. Every millimeter of built-in adjustability is a dimension that doesn't need to be measured perfectly. Outcome data (§2.3) calibrates, per dimension type, when the depth prior is trustworthy — over time, fewer questions, same fit rate.

2.3 The fit-outcome record (proprietary data)

Field	Example
intent_id / design_id / order_id	int_0912 / dsn_2210 / ord_1873
category + template + parameters	bracket.shelf_L · {span: 212mm, load_hint: books, screw: #8}
process / material / machine	FDM · PETG · prusa_mk4_02
dimension source per dim	user_measured · depth_inferred · assumed (+ confidence, + cross-check result)
outcome	fit_first_try | fit_after_reprint | returned | field_failure
evidence	customer photo in place; failure note; reprint delta (e.g., +0.4mm clearance)

This table is the company. It teaches which photo-inferred dimensions can be trusted, which clearances each material/machine needs, and which templates fail how — knowledge Zoo, Backflip, and Xometry structurally never collect because they never see the part arrive and fit (or not).

3 System Architecture — API-First

Everything is built as the **Vulcan Core API**; the consumer site is client #1, the founder's fulfillment dashboard is client #2, and future B2B customers are client #3. This forces clean seams and makes the fulfillment switch (founder printers → farms) a routing change, not a rewrite.

3.1 Core API surface (v0)

Endpoint	Purpose
POST /intents	Photo(s) + text → IntentSpec (category, candidate template, extracted dims + confidence, open questions)
POST /intents/{id}/answers	User answers to measurement prompts; updates IntentSpec
POST /designs	IntentSpec → geometry (Track A params or Track B mesh) + DFM report
GET /designs/{id}/preview	Render (PNG/GLB) + key dimension callouts

Endpoint	Purpose
POST /quotes	Design + material + destination → price, lead time
POST /orders	Quote acceptance + Stripe payment → queued for fulfillment
GET /orders/{id}	Status: queued → printing → QA → shipped → delivered
POST /orders/{id}/outcome	Fit-outcome record (customer flow or founder-entered)
GET /catalog/templates	Template library w/ parameter schemas (future B2B surface)

3.2 Components & stack

Component	Recommendation	Notes
API server	Python · FastAPI · Postgres · S3-compatible storage	Boring and fast to build alone
Intent parser	Frontier LLM (Claude) with strict JSON schema output	Photos + user annotations passed to vision model; emits IntentSpec incl. questions[]
Depth service	Depth Pro / Metric3D-class open model (self-hosted GPU or per-call inference API)	Metric-scale prior + unit-mistake cross-check (§2.2); never sole source for critical dims
Template engine	CadQuery or build123d (code-CAD, parametric, Python-native)	Each template = reviewed Python module + parameter schema + DFM rules
Freeform adapter	Zoo text-to-CAD API (swappable adapter interface)	Never load-bearing; always founder-reviewed in v0
DFM service	trimesh + manifold3d checks; PrusaSlicer CLI as printability + cost oracle	Slicer output (grams, minutes) feeds pricing directly
Pricing engine	Deterministic formula over slicer output (§5)	No ML needed at this stage
Web app	Next.js thin client on the API; Stripe Checkout	One flow: photo → questions → preview → pay
Fulfillment dashboard	Simple queue view; per-order gcode + QA card download	The founder is the fulfillment network in Phase 0

4 Fulfillment Phases

Phase	Who prints	Capacity	Materials	Gate to next phase
0 (mo. 1–4)	Founder's own printers	~5–15 orders/day depending on printer count and part size	PLA, PETG, TPU; CF-PETG if hardware allows	Demand proven: ≥50 paid orders/mo + fit-success ≥90%
1 (mo. 4–12)	Partner print farms via API (Slant 3D-class, JLC3DP-class) + founder for QA/exceptions	Hundreds/day; founder time shifts to templates + growth	Adds SLS/MJF nylon (PA12) for engineering parts	Farm QA parity: farm fit-success within 3 pts of founder's
2 (yr. 2)	Multi-process routing: farms + CNC + sheet-metal partners (SendCutSend-class); Xometry-class for metal one-offs	Effectively unbounded	Adds aluminum, steel, laser-cut/bent sheet	Unit economics hold at ≥60% gross margin blended

Phase 0 is deliberately founder-fulfilled: it is the cheapest possible demand test, it generates revenue immediately, and hand-printing every early order builds the DFM intuition that gets encoded into templates. It is a learning phase wearing a fulfillment costume — exit it as soon as the gate is met.

5 Unit Economics (Phase 0 worked example)

Line	Example: shelf bracket, PETG, 140g	Notes
Material	$3.10	~$22/kg PETG incl. waste factor 1.25
Machine time	$1.80	6h print · $0.30/h (power, wear, depreciation)
Founder labor	$5.00	~15 min: QA, pack, label (priced honestly at $20/h)
Packaging + shipping	$6.20	USPS Ground Advantage, boxed
Landed cost	**$16.10**	
Price	$39	Value-priced vs. replacement/frustration, not cost-plus
Gross margin	**~59%**	Target: ≥55% Phase 0, ≥60% blended Phase 1+

Pricing formula: **price = max(category floor, landed cost × 2.2)**, rounded to a $9-ending price point. Reprints for fit failures are free (they buy outcome data and word of mouth); cap at one free reprint per order.

6 The 90-Day Solo-Founder Build Plan

Weeks	Build	Exit criteria
1–2	Template engine + first 3 templates (shelf bracket, tube adapter, appliance knob) in CadQuery. Slicer-in-the-loop DFM + cost. Print each on own hardware; iterate tolerances.	3 templates print reliably; parameter → part in <10 min untouched

Weeks	Build	Exit criteria
3–4	Intent parser (Claude w/ JSON schema) + depth-model scale prior + guided-measure question flow w/ arrows drawn on the photo + unit cross-check. Core API v0 (intents, designs, quotes). Internal CLI ordering.	Photo + sentence → correct rendered part for 8/10 test cases; unit-mistake catch demonstrated
5–6	Web app v0 (photo → questions → preview → Stripe). Order queue + QA card. Ship 10 friendly-user orders end-to-end. Add 2 more templates from what they ask for.	10 real orders delivered; first fit-outcome rows collected
7–8	Soft launch in 2–3 obsessive niches (r/vandwellers, RV forums, appliance-repair groups, r/HomeImprovement). Founder reviews every design. Free reprint policy live.	50 paid orders; fit-first-try ≥80%; support <20 min/order
9–10	Fix the top 3 failure modes in templates. Add Track B (Zoo adapter) behind founder review for out-of-template requests. “We can't make that” logging + demand dashboard.	Track B serving long-tail orders; demand log ranks next categories
11–12	Outcome-capture flow w/ $5 photo credit. Weekly template releases driven by demand log. Talk to 2 print farms; run 5 duplicate orders through a farm to compare QA.	150 cumulative orders; ≥20% repeat/referral; farm parity data in hand

Total cash needed: roughly $2–5k (filament, shipping supplies, LLM/API credits, hosting, Stripe). The founder's printers and time are the real investment.

7 Metrics & Go / Kill Criteria (day 90)

Metric	Go	Kill / rethink
Fit success, first try	≥ 85% and rising	< 70% after template fixes — fit inference isn't working
Repeat or referral rate (60-day)	≥ 20%	< 8% — novelty purchases, not a habit
Orders/week trend (wk 8–12)	Growing without paid spend	Flat despite niche saturation
Gross margin	≥ 55%	< 40% — support/reprints eating the model
Support time per order	< 15 min and falling	> 30 min — doesn't scale past founder
Demand-log breadth	Requests cluster into templatizable categories	Requests are unclusterable one-offs

8 Top Risks & Mitigations

Risk	Mitigation
Backflip or Zoo partners with a marketplace and ships the full loop first	Speed to outcome data; own the consumer relationship and templates; be an attractive partner, not a clone
Fit failures poison early word of mouth	Track A templates only at launch; confirm every critical dim; free reprint; publish fit-rate honestly
Founder fulfillment becomes the bottleneck before demand is proven	Hard cap daily orders (waitlist beats bad QA); exit to farms at the Phase 0 gate, not later

Risk	Mitigation
Liability from a part that fails in use	Category exclusions (§1.4), load disclaimers at checkout, LLC + product liability insurance before public launch
Text-to-geometry commoditizes into free phone features	The moat was never generation — it's fit-outcome data, templates, and fulfillment QA; double down there

9 Beyond Day 90 (headline only)

- **Months 4–12:** farm routing live; 15–25 templates; SLS nylon tier; the "make it fit" play — pair commodity products with custom interface parts (§the TV insight); first B2B API pilot (repair shops, RV outfitters).
- **Year 2:** multi-process routing (CNC, sheet metal); template marketplace for third-party designers (rev share); fine-tune generation on accumulated outcome data — the point where Vulcan's designs fit better than raw text-to-CAD ever can.
- **The compounding claim:** every order makes the next order more likely to fit on the first try. No competitor starting later can buy that.

Appendix A — Template Library v1 parameter sketches

Template	Core parameters (LLM-filled, user-confirmed)
bracket.shelf_L	span, depth, thickness, load_hint → rib count, screw size/count, wall offset
adapter.tube	od_a, id_a, od_b, id_b, engagement length, taper, thread? (printed threads ≥ M8 only)
knob.appliance	shaft type (D/round/spline), shaft dia, depth, knob dia, grip style, pointer?
enclosure.box	inner l/w/h, wall, lid type (snap/screw), port cutouts[], standoffs[]
hook.rail	rail profile (photo-matched), load_hint, reach, tip style
clip.cable	bundle dia, mount (screw/adhesive/magnet), count

Each template ships with: parameter schema + validation ranges, DFM rules (min wall, orientation, clearance per material), QA card spec, and 3 golden test prints on file.